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Intelligent Aquaculture: Leveraging IoT and AI for Sustainable Shrimp Farming in Sri Lanka

Get Started

Our Supervisors

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With a growing population comes
a growing need to feed the world.

AI-Agentic Farm Management Assistant

Specialization: B.Sc(Hons) Information Technology(SLIIT)

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AI-Agentic Farm Management Assistant

Existing Gaps in Shrimp Farming

Knowledge Gaps /Problems

- Labor & Resource Allocation Issues
- Poor Benchmarking Against Standards
- Limited Real-Time Monitoring

Shortcoming of the existing Systems

- No performance tracking → KPI benchmarking vs standard
- Poor Resource Optimization → Optomization Engine
- Lack of Data-Driven decesion making→ Decision Support Dashboard

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AI-Agentic Farm Management Assistant

Proposed Solutions

Optimization Engine

Recommends best feeding, labor, and harvest plans.

KPI Benchmarking

Tracks FCR, energy use, and yield against standards to spot gaps.

Decision Support Dashboard

dashboard with real-time insights and alerts.

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AI-Agentic Farm Management Assistant

Domain Knowledge Applied

- Ai Agents
- Data Science & Analytics
- Human-Computer Interaction

Technologies Used

- Crew Ai
- Scikit-learn
- ReactJS
- AWS
- MongoDB
- Grafana

Evaluation & Success

- Benchmarking: Use industry standards and existing automated systems for comparison.
- Farmer Feedback: Gather user satisfaction and ease-of-use insights from farm operators.

Data & Ethics

Data Sources: IoT sensor data (water quality), historical farm records (feed, labor, yield), and disease

Ethics: Farmers' personal and farm data anonymized before storage.

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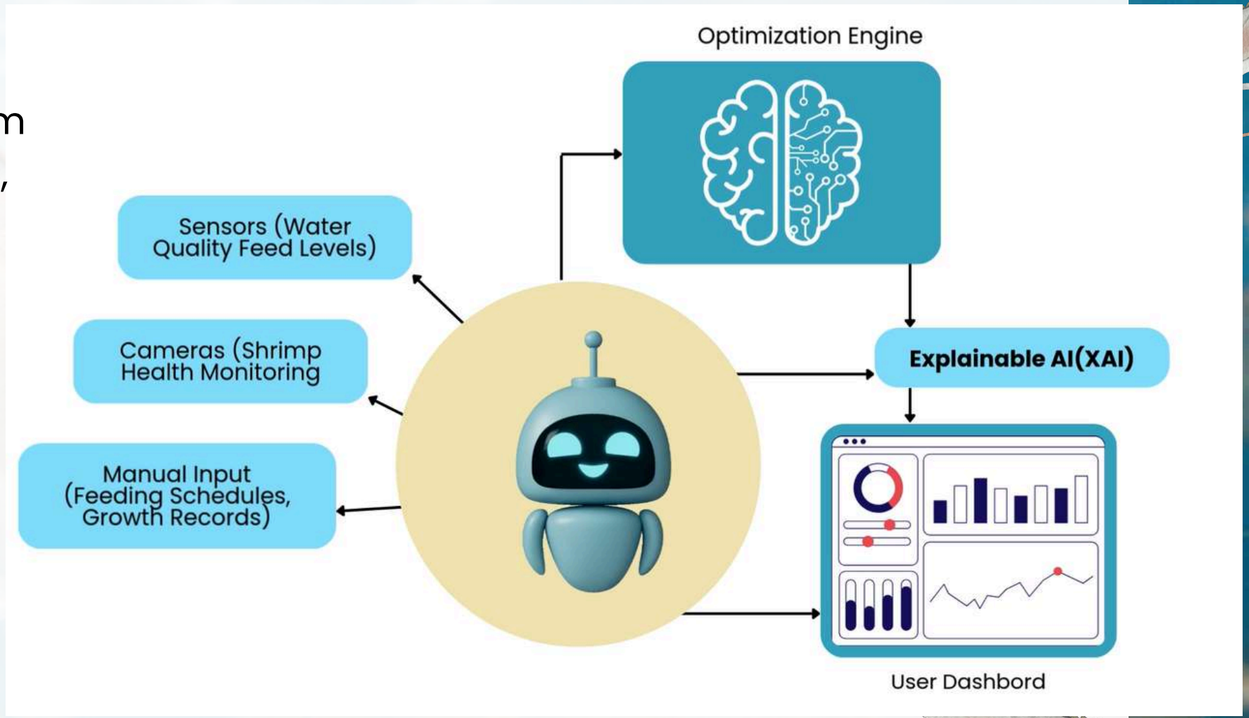
AI-Agentic Farm Management Assistant

Real-World Scenario

- If dissolved oxygen is predicted to drop at night, the system suggests turning on aerators and adjusting feed amounts, saving both feed and energy costs.

- A farmer checking the app in the morning sees today's feeding plan, staff allocation, and an alert about possible water quality risks, enabling quick and informed action.



An AI-Powered Monitoring and Disease Detection System

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An AI-Powered Monitoring and Disease Detection System

Manual Limitations

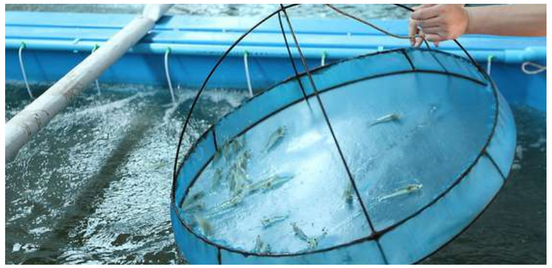
- **Manual Monitoring**: Time-consuming, error-prone, and lacks real-time insights.
- **Delayed Disease Detection**: Leads to high shrimp mortality and economic loss.

AI-Powered Precision

- **AI-powered System:** Real-time shrimp behavior monitoring and early disease detection using computer vision and IoT sensors.

Knowledge Gap

- No real-time monitoring of shrimp behavior.
- Inadequate early disease detection, leading to poor farm management.



Shortcomings of Existing Systems

- Manual Monitoring
- Labor-intensive
- Delayed Diagnosis
- financial lost

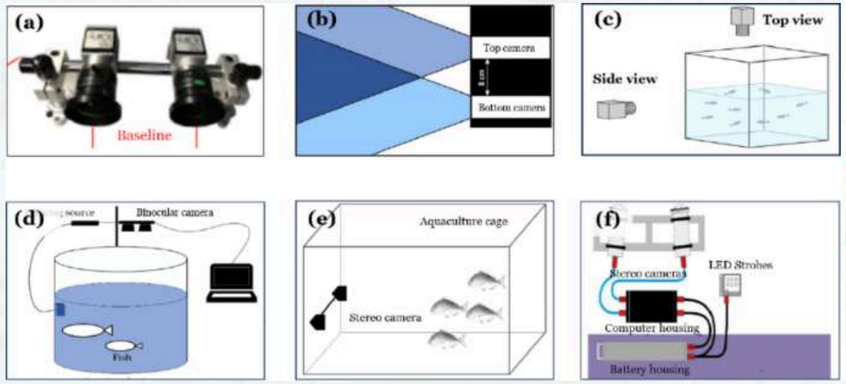
Proposed Solution

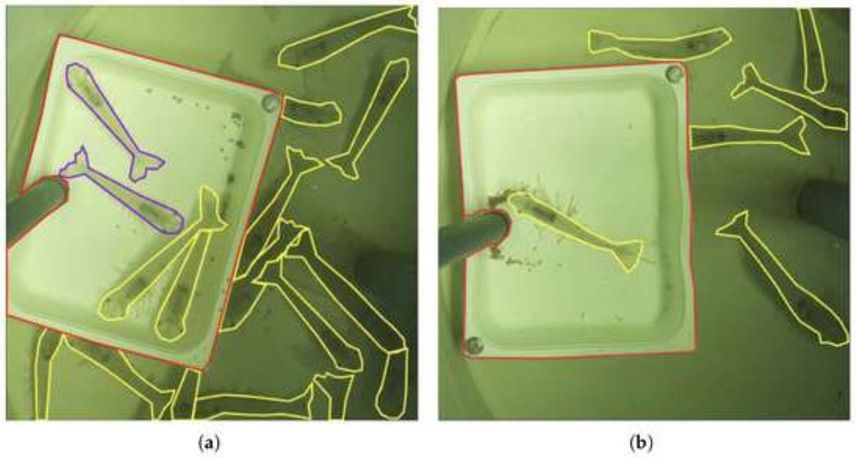
- AI-Powered Monitoring
- Data Integration
- Automated Alerts

An AI-Powered Monitoring and Disease Detection System

Application of Specialization-based Knowledge

- **Computer Vision**: Used for behavioral analysis through camera imagery.
- **Machine Learning**: Applied for behavioral classification and disease recognition.
- **Data Science/Environmental Science**: Used to correlate environmental conditions with shrimp behavior.
- **IoT**: Integrating environmental sensor data in real-time.

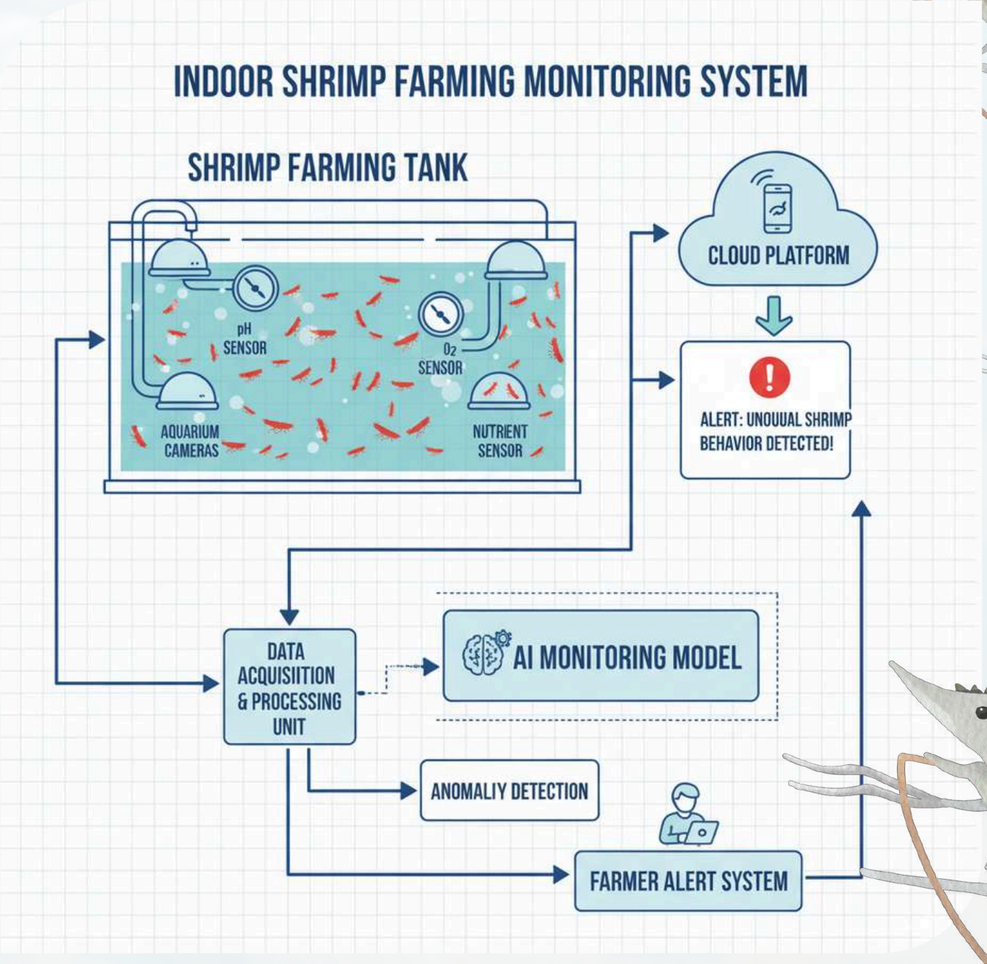




An AI-Powered Monitoring and Disease Detection System

Evaluation Metrics

- **Accuracy**: Aim for 90%+ accuracy in detecting disease and abnormal behavior.
- **Validation**: Use traditional diagnostic methods for comparison, such as visual inspection by farmers.
- **Field Testing**: Trial in real shrimp farms for operational performance and farmer feedback.



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An AI-Powered Monitoring and Disease Detection System

Real-World Scenario

1	**At 2:15 AM**, the system notices the dissolved oxygen (DO) level drops from 5.6 to 3.2 mg/L, and shrimp start clustering at the edges and stop eating. This indicates a Warning signal.
2	**By 6:40 AM**, the system detects white feces, a sign of potential disease, and sends an alert to the farmer via SMS/WhatsApp.
3	**Actions Taken:** • The farmer follows the dashboard suggestions: turns on aerators, exchanges water, and reduces feeding. • The system tracks the improvement in DO levels and updates the risk level back to "Normal" as conditions stabilize.
4	**Outcome** • :The shrimp recover within 48 hours, preventing a disease outbreak and avoiding large-scale mortality.

BIOFLOC AUTOMATED FEED MANAGEMENT SYSTEM

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BIOFLOC AUTOMATED FEED MANAGEMENT SYSTEM

Automated Feed Management via Acoustic and Visual Sensing

The Problem: Overfeeding

Traditional shrimp farming often relies on manual, time-based feeding, leading to significant overfeeding (up to 30% waste). This practice not only inflates production costs but also severely pollutes the water.

Consequences: Waste & Pollution

Excess feed decomposes, degrading water quality and harming the shrimp environment. Manual methods are inefficient, and existing automated systems lack the intelligence to adapt to real-time conditions.

Our Solution: Intelligent Sensing

We introduce a novel system using a hydrophone and camera to create a real-time feedback loop. This intelligent approach ensures precise feeding, minimizes waste, and optimizes shrimp health.

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BIOFLOC AUTOMATED FEED MANAGEMENT SYSTEM

Conventional vs. AI-Powered Automated Shrimp Feeding Systems

Feature	Manual/Timer Systems	Automated & Intelligent System
Feeding	Person feeds by hand or simple machine on fixed schedule	Smart system uses camera and hydrophone to feed only when needed
Technology	Basic or no technology	IoT and AI sensors
Cost	Higher costs from waste	Saves money
Effort	Farmer must check multiple times daily	Works automatically
Results	Food waste and slower growth	30% less waste, 20-30% faster growth

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BIOFLOC AUTOMATED FEED MANAGEMENT SYSTEM

Application of Specialisation- Based knowledge

Key Domains

- **Internet of Things (IoT):** Integrates hardware for real-time data collection.
- **AI/ML**: Edge AI processes audio/visual data for automated feeding decisions.
- **Automation**: Stepper motor and hopper automate precise feed distribution.
- **Signal Processing**: Analyzes hydrophone data to detect shrimp feeding clicks.

Latest Technologies

- **Edge Impulse**: Trains and deploys ML models for real-time, offline data analysis.
- **YOLOv8**: Identifies "uneaten feed" and "shrimp activity" with speed and accuracy
- **Arduino/ESP32-CAM**: Low-cost microcontrollers for IoT and visual sensing.

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BIOFLOC AUTOMATED FEED MANAGEMENT SYSTEM

How to Evaluate Success

Validation: We will prove our system works by testing it against a traditional manual feeding method. We will measure three things: how much feed is used for the same amount of shrimp growth, how fast the shrimp grow, and how clean the water stays.

Data & Ethics

- Farmer consent
- Non-invasive monitoring
- Local storage of data
- fair compensation for farmers

90%

Feeding Decision Accuracy

Target accuracy for identifying shrimp hunger and satiation.

30%

Reduced Feed Waste

Expected reduction in feed waste through precise automation.

20-30%

Improved Shrimp Growth

Anticipated improvement in shrimp growth rates.

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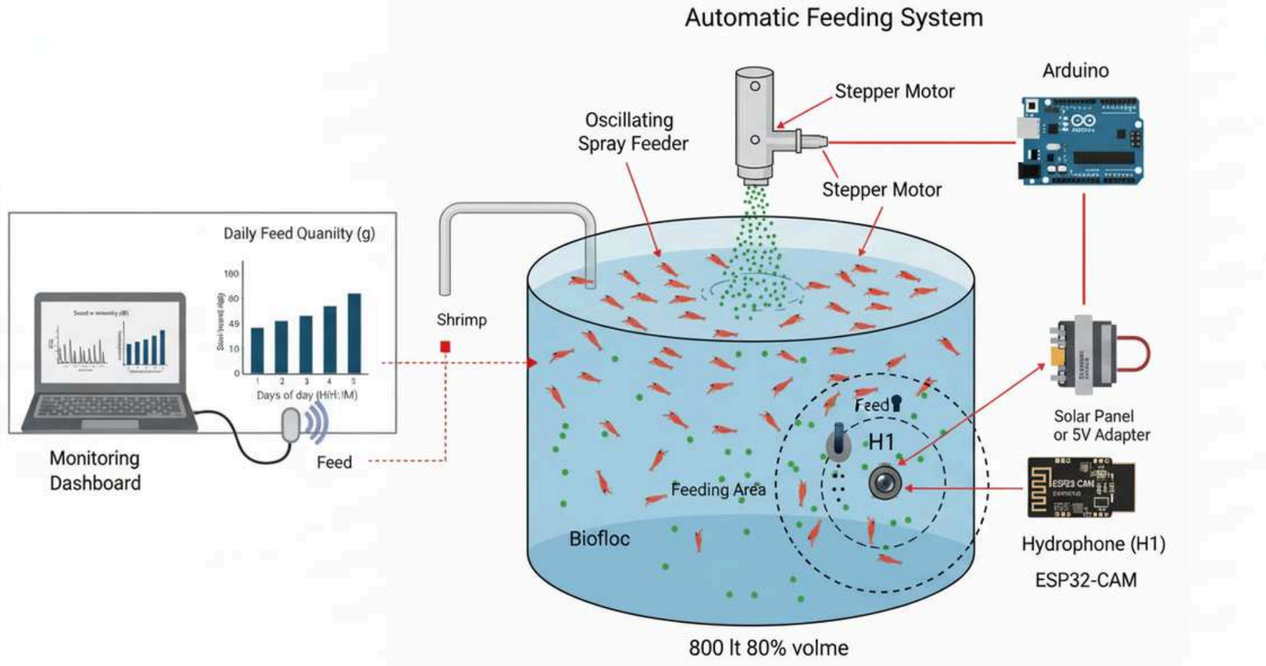
BIOFLOC AUTOMATED FEED MANAGEMENT SYSTEM

System Architeture



How to use it in a real-world scenario

- Farmers mount the system's frame above their indoor biofloc tank, placing sensors in the feeding area.
- After initial setup, the system operates autonomously, running feeding cycles based on real-time shrimp needs.
- Farmers only need to refill the hopper and perform routine maintenance. The system logs all feeding events, providing data on consumption and waste.

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Water Quality Monitoring in Shrimp Farming

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The Pitfalls of Traditional Monitoring

Knowledge Gap

- Shrimp farming in Sri Lanka suffers losses due to poor water quality.
- Manual monitoring is slow and error-prone.
- Lacks real-time data on key parameters (e.g., Dissolved Oxygen, Ammonia).
- Farmers react after problems occur, not proactively.

Shortcomings of Existing Systems

- Current systems rely on manual labor and fixed schedules.
- Ammonia is tested weekly in labs, with 24-hour result delays.
- No real-time monitoring or automated control.
- Critical water quality changes may go unnoticed, harming shrimp.

Proposed Solution

- IoT-based system monitors DO, Ammonia, and other parameters.
- Microcontroller automates aerators, pumps, and dosing systems.
- Cloud dashboard enables remote monitoring and real-time alerts.
- Supports instant response, reduces shrimp mortality, and boosts productivity.

Key Domains and Emerging Technologies in Smart Shrimp Farming

Key Domains:

- Internet of Things (IoT): Real-time data collection with sensors and microcontrollers.
- AI/ML: Edge AI analyzes real-time and historical data and environmental inputs, for forecasting and decision-making.
- Automation: Controls aerators, pumps, and chemical dissolvers with precision.
- Signal Processing: Uses PH Detector DO Detector to process signals

Latest Technology :

- Edge Impulse → Runs ML models on sensor data (DO, pH, ammonia) for real-time analysis.
- YOLOv8 → Used for fast anomaly detection in sensor readings to spot water quality issues quickly.
- Arduino / ESP32 → Collects data from sensors and triggers automatic aerators/pumps via IoT.

BIOFLOC AUTOMATED FEED Water Quality Monitoring in Shrimp FarmingMANAGEMENT SYSTEM

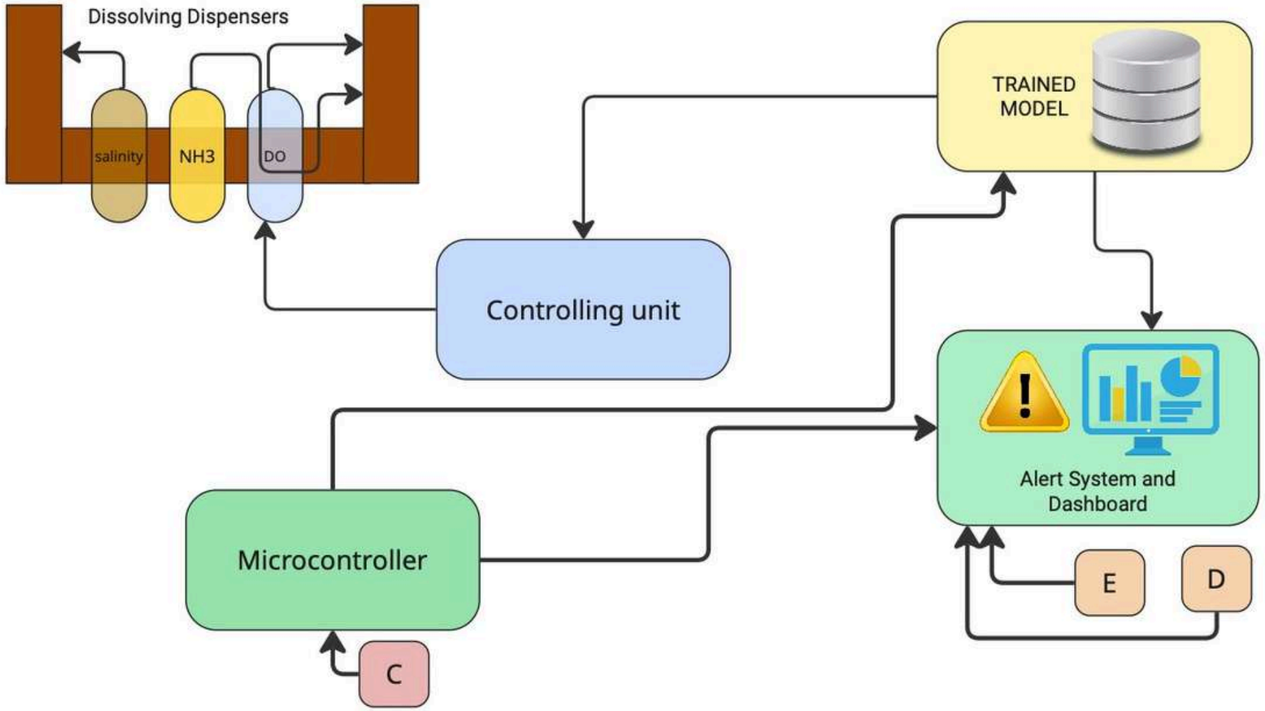
How to Evaluate Success :

- Measure water quality stability (DO, pH, ammonia) compared with manual monitoring.
- Compare accuracy and reliability with traditional manual testing.
- Track system response time in automatically controlling aerators and pumps.
- Assess cost savings from reduced labor and fewer chemical treatments.
- Evaluate overall consistency in maintaining optimal water quality.

Data & Ethics

- Ensure sensor calibration and regular validation for accurate readings.
- Protect data with secure storage, encryption, and controlled access.
- Keep systems transparent and easy for farmers to use.
- Maintain affordability and scalability so even small farms benefit.
- Promote sustainability by reducing water pollution and chemical waste.

Completion of the project



- Sensors monitor DO, ammonia, temperature, and salinity in the shrimp pond.
- If DO drops below 5 mg/L, the microcontroller instantly switches on the aerator.
- When ammonia rises, the system activates the pump to refresh water.
- Data is sent to a cloud dashboard for live monitoring and alerts.
- This ensures real-time control, healthier shrimp, and higher farm efficiency.

2024 Presentation

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Intelligent Aquaculture: Leveraging IoT and AI for Sustainable Shrimp Farming in Sri Lanka

COMMERCIALIZATION

- **First** integrated IoT-AI solution for shrimp disease prevention in Sri Lanka
- **Solves** critical 90% farm failure rate in Puttalam district due to EHP/WSSV diseases
- **Partnerships** established with University of Ruhuna & NAQDA for validation
- **High scalability** potential across 2,500+ active shrimp farms nationwide
- **Multiple revenue streams** through hardware sales + SaaS monitoring subscriptions

COST

- IoT sensors (pH, DO, temperature, salinity), ESP32 microcontroller, GSM modules
- Smartphone microscopy attachments, automated feeder mechanisms
- Cloud infrastructure, mobile app development, biofloc tank setup

Estimated Initial Investment: Rs. **30,000-45,000**(prototype development + pilot testing)

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COMMERCIAL MODEL & MARKET

Cost Recovery/Pricing

- Hardware Sales - Complete IoT monitoring kit (Rs. 45,000) with 33% profit margin
- SaaS Subscriptions - Monthly monitoring service (Rs. 5,000/month per farm)
- Premium Services - Disease detection, feeding optimization, consultation (Rs. 8,000/setup)

Target Market & Customer Profile

- Primary - Small-scale shrimp farmers in Puttalam/Northwestern provinces (2,500+ farms)
- Secondary - Medium commercial aquaculture operations requiring automated monitoring
- Tertiary - Government cooperatives and NAQDA-supported farmer groups

Value Proposition

- Disease Prevention - Early EHP detection (7-10 days) prevents 80-100% mortality losses
- Feed Optimization - AI-driven feeding reduces 60-70% of production costs through FCR improvement
- Real-time Monitoring - 24/7 water quality tracking replaces manual testing, saves 4+ hours daily
- Profit Increase - 40% higher yields through optimal environmental conditions

Thank You